

Playbook:

Enabling Enhanced Logging in Windows for Wazuh (with Sysmon)

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1. Objective

- To enable, configure, and validate enhanced logging on Windows endpoints for collection by the Wazuh platform.
- Enhanced logging increases visibility into endpoint activity, enabling improved threat detection, investigation, and compliance reporting.

2. Scope

This playbook applies to:

- Windows 10, Windows 11, and Windows Server endpoints.
- Wazuh agents are installed and registered to a Wazuh Manager.
- Enhanced logging components: Sysmon, PowerShell Operational, Windows Security Events, and Windows Defender logs.

3. Prerequisites

- Windows 10/11 or Windows Server endpoint.
- Administrator privileges on the machine.
- Wazuh Manager is accessible on the network.
- Wazuh agent is installed and connected to the manager.
- If not installed, follow the [Wazuh Agent Deployment Playbook](#) before proceeding.
- Internet access to download Sysmon binaries and configuration.

4. Procedure

4.1 Phase 1 — Deploy Sysmon

Action:

1. Download Sysmon from Microsoft Sysinternals:
<https://learn.microsoft.com/en-us/sysinternals/downloads/sysmon>
2. Extract binaries to C:\Tools\Sysmon.
3. Obtain and use a maintained Sysmon configuration (e.g., [SwiftOnSecurity sysmon-config](#)).
4. Install Sysmon with the configuration in PowerShell:

```
cd C:\Tools\Sysmon
```

```
.\Sysmon64.exe -accepteula -i .\sysmonconfig-export.xml
```

Validation:

- Service check: Get-Service Sysmon64 → Status = Running.
- Event Viewer → Applications and Services Logs → Microsoft → Windows → Sysmon → Operational.
- Confirm events for Process Create (ID 1) and Network Connect (ID 3) are logged.
- The default Sysmon configuration logs all events, which can be noisy. Use a community-maintained config such as SwiftOnSecurity's for balanced visibility and performance.

4.2 Phase 2 — Enable Additional Windows Logging

Action:

Open Local Group Policy Editor (gpedit.msc).

Configure PowerShell logging:

Path: Administrative Templates → Windows Components → Windows PowerShell.

Enable:

- Turn on Module Logging
- Turn on Script Block Logging
- Turn on PowerShell Transcription

Enable Security auditing:

```
auditpol /set /subcategory:"Process Creation" /success:enable /failure:enable
```

```
auditpol /set /subcategory:"Logon" /success:enable /failure:enable
```

```
auditpol /set /subcategory:"Policy Change" /success:enable /failure:enable
```

Validation:

Event Viewer → Confirm logs appear under:

- Microsoft-Windows-PowerShell/Operational
- Security
- Microsoft-Windows-Defender/Operational

4.3 Phase 3 — Configure Wazuh Agent

Action:

Edit C:\Program Files (x86)\ossec-agent\ossec.conf.

Add the following <localfile> entries:

```
<!-- Sysmon Logs -->
```

```
<localfile>
```

```
<location>Microsoft-Windows-Sysmon/Operational</location>
```

```
<log_format>eventchannel</log_format>
```

```
</localfile>
```

```
<!-- PowerShell Logs -->
```

```
<localfile>
```

```
<location>Microsoft-Windows-PowerShell/Operational</location>
```

```
<log_format>eventchannel</log_format>
```

```
</localfile>
```

```
<!-- Windows Defender Logs -->
```

```
<localfile>
```

```
<location>Microsoft-Windows-Windows Defender/Operational</location>
```

```
<log_format>eventchannel</log_format>
```

```
</localfile>
```

```
<!-- Windows Security Logs -->
```

```
<localfile>
```

```
<location>Security</location>
```

```
<log_format>eventchannel</log_format>
```

```
</localfile>
```

Save the configuration and restart the Wazuh agent, run the command in PowerShell:

Restart-Service wazuh

Validation:

- Review ossec.log at C:\Program Files (x86)\ossec-agent\ossec.log.
- Confirm entries such as:
- wazuh-agent: INFO: (1950): Analyzing event log: 'Microsoft-Windows-Sysmon/Operational'

4.4 Phase 4 — Validate End-to-End Log Ingestion

Action:

Trigger test events:

Open notepad.exe → Expect Sysmon Event ID 1.

Run ping 8.8.8.8 → Expect Sysmon Event ID 3.

Execute a PowerShell script → Expect PowerShell logging.

On Wazuh Manager:

tail -f /var/ossec/logs/alerts/alerts.json | grep Sysmon

In Wazuh Dashboard:

Navigate to Security Events.

Search/filter for:

provider_name: Microsoft-Windows-Sysmon

rule.groups: sysmon

Validation:

- Sysmon, PowerShell, Defender, and Security events appear in Wazuh alerts.
- Analysts can query and pivot on enhanced logs in the Wazuh dashboard.

5. Troubleshooting

- Agent does not start: Validate XML syntax in ossec.conf.
- No Sysmon events in Wazuh: Confirm Sysmon logs exist in Event Viewer.

- Excessive noise: Tune Sysmon config XML or use XPath filters in Wazuh.
- Permission denied saving ossec.conf: Run Notepad as Administrator.
- Logs not visible in dashboard: Check Filebeat/Indexer status.
- Events missing from the Security log: Run auditpol /get /subcategory:* to confirm that auditing is enabled.
- Sysmon service not running: Verify installation with sc query sysmon64 and reinstall if necessary.

6. References

[Wazuh Documentation – Windows Logs](#)

[Sysmon \(Microsoft Sysinternals\)](#)

[SwiftOnSecurity Sysmon Config](#)

7. Expected Outcome

After following this playbook:

- Sysmon and enhanced Windows logs are enabled.
- Wazuh agent collects logs from Sysmon, PowerShell, Defender, and Security channels.
- Events are visible in both alerts.json and the Wazuh dashboard.
- SOC analysts gain deeper visibility into process creation, network connections, PowerShell activity, and Defender alerts. This improves detection of common attack techniques (e.g., malware execution, persistence).

Enhanced logs provide rich telemetry for detecting techniques such as process injection, credential dumping, suspicious PowerShell execution, and persistence via registry changes.